AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 7
Differential Equations

100-Question Master Practice Set

- Questions 1–90: Multiple Choice
- Questions 91–100: Free Response
- Complete answer key, worked solutions, and scoring rubrics appear after Question 100

Exam Structure & Priority Guide

Questions are continuously numbered from 1 to 100. Answers and solutions begin only after Question 100.

Priority Ratings

★★★★★	**Very Important**	Slope fields, separable equations, exponential models, and logistic behavior.
★★★	**Mandatory**	Verifying solutions, equilibrium solutions, initial-value problems, and qualitative reasoning.
★★	**Important**	Basic model interpretation, units, and direct slope calculations.

- **Q1–Q90:** MCQ with four options and one correct answer.
- **Q91–Q100:** Multi-part FRQ, each worth 9 points; graph and table stimuli included.
- **Unit scope:** modeling, solution verification, slope fields, separable equations, exponential and logistic models.
- **Calculator:** not required for the intended methods; exact expressions are accepted.

AP Calculus AB Unit 7 Topic Coverage

7.1 Modeling Situations with DEs

7.2 Verifying Solutions

7.3 Sketching Slope Fields

7.4 Reasoning from Slope Fields

7.5 Separation of Variables

7.6 General Solutions

7.7 Exponential Models

7.8 Logistic Models

> **Directions:** Attempt the entire paper before checking answers. For FRQs, show separation steps, use initial conditions, and state units when appropriate.

QUESTION 01

EASY · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A population P grows continuously at 4% per year. Which differential equation models this?

A $P=0.04t$

B $P'=0.04P$

C $P'=4P$

D $P'=0.04$

QUESTION 02

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

An object cools in a room held at 20°C. Newton's cooling model is

A $T'=kT$

B $T'=k(T+20)$

C $T'=-20k$

D $T'=-k(T-20)$, $k>0$

QUESTION 03

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If a population satisfies $P'/P=0.03$, then its differential equation is

A $P'=0.03$

B $P'=P/0.03$

C $P'=0.03P$

D $P=0.03t$

QUESTION 04

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A quantity A receives a constant input of 12 units per hour and is removed at a rate proportional to A, with proportionality constant 0.2. The model is

A $A'=12A-0.2$

B $A'=12+0.2A$

C $A'=12-0.2A$

D $A'=0.2A-12$

QUESTION 05

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $P'=0.4P(1-P/1000)$, the carrying capacity is

A 0.4

B 0

C 400

D 1000

QUESTION 06

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The equilibrium solutions of $y'=y(5-y)$ are

A $y=0$ and $y=5$

B $y=5$ only

C $y=-5$ and $y=0$

D $y=1$ and $y=5$

QUESTION 07

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y' = 3 - y$, the equilibrium $y = 3$ is

A unstable
B not an equilibrium
C stable
D neither stable nor unstable

QUESTION 08

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $P' = 0.2P(1 - P/50)$, if $P = 60$ then P' is

A undefined
B zero
C positive
D negative

QUESTION 09

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

In $P' = kP$, where t is measured in years, the units of k are

A people
B years
C per year
D people per year

QUESTION 10

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $P' = 0.5P(1 - P/500)$, the growth rate is zero when $P =$

A 0 or 500

B 0 only

C 500 only

D 250

QUESTION 11

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Which function solves $y' = 2y$?

A $y = 2Ce^x$

B $y = Ce^x$

C $y = Ce^{2x}$

D $y = C + 2x$

QUESTION 12

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Which function solves $y' = 2x$?

A $y = x^2/2 + C$

B $y = 2x^2 + C$

C $y = x^2 + C$

D $y = x + C$

QUESTION 13

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Which function solves $y' = -y$?

A $y = 5 - t$

B $y = 5e^{-t}$

C $y = 5e^{t}$

D $y = e^{-t} + 5$

QUESTION 14

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The general solution of $y' = -3(y - 2)$ is

A $y = 2 - 3x$

B $y = 2 + Ce^{-3x}$

C $y = 2 + Ce^{3x}$

D $y = Ce^{-3x}$

QUESTION 15

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Which constant function is a solution of $y' = y(5 - y)$?

A $y = 1$

B $y = 0$

C $y = -1$

D $y = 5/2$

QUESTION 16

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The function $y=4/(1+3e^{-4t})$ solves

A $y'=y(1-y/4)$

B $y'=4y$

C $y'=-4y$

D $y'=y(4-y)$

QUESTION 17

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Which function solves $y'=1/x$ for $x>0$?

A $y=1/x+C$

B $y=\ln x+C$

C $y=x+C$

D $y=e^x+C$

QUESTION 18

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Which function solves $y'=\cos x$?

A $y=\sin x+C$

B $y=-\sin x+C$

C $y=-\cos x+C$

D $y=\cos x+C$

QUESTION 19

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Which function solves $y'=2xy$?

A $y=Ce^{2x}$

B $y=Ce^{x2}$

C $y=Ce^{x}$

D $y=Cx^{2}$

QUESTION 20

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Which family solves $y'=3-y$?

A $y=3t+C$

B $y=Ce^{-t}$

C $y=3+Ce^{t}$

D $y=3+Ce^{-t}$

QUESTION 21

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y'=x+y$, the slope at $(2,-1)$ is

A 1

B 3

C -3

D -1

QUESTION 22

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = 2x - y$, the slope at $(1,3)$ is

A -1

B 5

C 3

D 1

QUESTION 23

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = y(4 - y)$, the slope along the row $y = 2$ is

A 0

B 2

C 4

D -4

QUESTION 24

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = 3 - x$, the slope field has horizontal segments along

A $x = 0$

B $y = 3$

C $x = 3$

D $y = 0$

QUESTION 25

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = 2 - y$, horizontal slope segments occur when

A $x = y$

B $x = 2$

C $y = 2$

D $y = 0$

QUESTION 26

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y' = (x-1)(y+2)$, the slope is zero when

A $x = 1$ or $y = -2$

B $x = y$

C only at $(1, -2)$

D $x = -1$ or $y = 2$

QUESTION 27

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = x^2 - 9$, the zero-slope vertical lines are

A $x = 0$ only

B $x = -3$ and $x = 3$

C $y = -3$ and $y = 3$

D $x = 3$ only

QUESTION 28

EASY · MULTIPLE CHOICE ★★ IMPORTANT

For $y' = x - y$, the slope at $(2,2)$ is

A 4

B -4

C 0

D 2

QUESTION 29

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = 1 + y^2$, which statement is true about the slope field?

A Slopes are negative for $y < 0$

B Slopes are zero on $y = 0$

C Slopes are positive at every point

D Slopes depend only on x

QUESTION 30

EASY · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = -y$, the slope along $y = 0$ is

A 1

B -1

C depends on x

D 0

QUESTION 31

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y'=0.5y(1-y/20)$, when $y=10$ a solution is

A increasing
B constant
C undefined
D decreasing

QUESTION 32

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y'=0.5y(1-y/20)$, when $y=25$ the solution is

A constant
B increasing
C undefined
D decreasing

QUESTION 33

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The equilibrium solutions of $y'=0.5y(1-y/20)$ are

A $y=20$ only
B $y=0$ only
C $y=10$ and $y=20$
D $y=0$ and $y=20$

QUESTION 34

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y' = -(y-4)$, the equilibrium $y=4$ is

A not an equilibrium
B a solution only if $x=0$
C stable
D unstable

QUESTION 35

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = y-4$, the equilibrium $y=4$ is

A unstable
B a local maximum
C not an equilibrium
D stable

QUESTION 36

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A solution of $y' = y(5-y)$ with $y(0)=0$ remains

A decreasing without bound
B equal to 0
C equal to 5
D increasing without bound

QUESTION 37

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y' = y(5-y)$ with 0

A decreases toward 0

B increases without bound

C increases toward 5

D is constant

QUESTION 38

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = y(5-y)$ with $y(0)=7$, the solution initially

A decreases toward 5

B remains constant

C increases without bound

D increases toward 5

QUESTION 39

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y' = -2y$ and $y(0)>0$, the solution

A decreases through 0 and becomes negative

B decreases toward 0

C is constant

D increases without bound

QUESTION 40

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y' = y^2$ at $y = -2$, the slope is

A -2
B 0
C -4
D 4

QUESTION 41

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A solution satisfies $y' = x$ and $y(0) = 1$. What is $y(2)$?

A 2
B 3
C 4
D 5

QUESTION 42

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A solution satisfies $y' = 2x$ and $y(1) = 3$. What is $y(3)$?

A 11
B 10
C 12
D 8

QUESTION 43

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A solution satisfies $y'=\cos x$ and $y(0)=1$. What is $y(\pi)$?

A π

B 1

C 2

D 0

QUESTION 44

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y'=(y-2)(y+1)$, which equilibrium is stable?

A $y=-1$

B $y=2$

C both

D neither

QUESTION 45

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The general solution of $dy/dx=6x$ is

A $y=6x^2+C$

B $y=3x^2+C$

C $y=3x+C$

D $y=x^6+C$

QUESTION 46

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The general solution of $y' = 2xy$ is

A $y = Ce^{2x}$

B $y = Ce^{x2}$

C $y = Ce^{x}$

D $y = Cx^{2}$

QUESTION 47

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

On an interval not containing $x = 0$, the general solution of $y' = y/x$ is

A $y = Cx^{2}$

B $y = C/x$

C $y = Ce^{x}$

D $y = Cx$

QUESTION 48

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The general implicit solution of $y' = x/y$ is

A $y^{2} = x^{2} + C$

B $y^{2} = 2x + C$

C $y = x^{2} + C$

D $y^{2} = -x^{2} + C$

QUESTION 49

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The general solution of $y'=(x+1)/(y+1)$ is

A $y^2=x+1+C$

B $y+1=x^2/2+x+C$

C $y^2/2+y=x^2/2+x+C$

D $y^2/2+y=x^2+x+C$

QUESTION 50

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The general implicit solution of $y'=e^{x-y}$ is

A $y=e^x+C$

B $e^{-y}=e^x+C$

C $y=e^x-x+C$

D $e^y=e^x+C$

QUESTION 51

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A separated solution of $y'=x^2/y^2$ satisfies

A $y^3=3x^2+C$

B $y^3=x^3+C$

C $y^2=x^3+C$

D $y=x^3+C$

QUESTION 52

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A general solution of $y' = 1 + y^2$ is

A. $\tan y = x + C$
B. $\ln(1 + y^2) = x + C$
C. $y = \tan x + C$
D. $\arctan y = x + C$

QUESTION 53

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The general nonzero solution of $y' = y/(1 + x^2)$ is

A. $y = C \arctan x$
B. $y = C(1 + x^2)$
C. $y = Ce^{\wedge}x$
D. $y = C\, e^{\wedge}(\arctan x)$

QUESTION 54

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

The general implicit solution of $y' = \cos x / y$ is

A. $y^2 = 2\sin x + C$
B. $y^2 = \sin x + C$
C. $y^2 = -2\sin x + C$
D. $y = 2\cos x + C$

QUESTION 55

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $y>0$, the separated general solution of $y'=(x+1)/\sqrt{y}$ can be written

A $2\sqrt{y}=x^2/2+x+C$

B $y^{(3/2)}=x^2+x+C$

C $(2/3)y^{(1/2)}=x^2/2+x+C$

D $(2/3)y^{(3/2)}=x^2/2+x+C$

QUESTION 56

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

On $x>-1$, the general nonzero solution of $y'=y/(x+1)$ is

A $y=C(x+1)^2$

B $y=Ce^{(x+1)}$

C $y=C/(x+1)$

D $y=C(x+1)$

QUESTION 57

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The nonzero solutions of $y'=y^2$ can be written

A $y=1/(x+C)$

B $y=Ce^x$

C $y=-(x+C)$

D $y=-1/(x+C)$

QUESTION 58

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The solution of $y'=y(1-y/4)$, $y(0)=1$, is

A $4/(1+3e^{-x})$

B $4/(1-3e^{-x})$

C $1/(1+3e^{-x})$

D $4/(1+e^{-x})$

QUESTION 59

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For the IVP $y'=x/y$, $y(0)=2$ and $y>0$, the solution satisfies

A $y^2=2x+4$

B $y^2=x^2+4$

C $y^2=x^2+2$

D $y=x^2+2$

QUESTION 60

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $y'=2x/(1+y)$, $y(0)=1$, the implicit solution is

A $y+y^2=x^2+3/2$

B $y+y^2/2=x^2+3/2$

C $y+y^2/2=x^2+1$

D $y^2/2=x^2+3/2$

QUESTION 61

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Solve $y' = 2xy$ with $y(0) = 3$.

A $3e^{x2}$

B $e^{x2}+3$

C $3e^{2x}$

D $3x^2$

QUESTION 62

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Solve $y' = -0.5y$, $y(0) = 80$.

A $80e^{-0.5t}$

B $80-0.5t$

C $80e^{0.5t}$

D $40e^{-t}$

QUESTION 63

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Solve $y' = 4-y$, $y(0) = 1$.

A $4-3e^{-t}$

B $1+4e^{-t}$

C $4+3e^{-t}$

D $3e^{-t}$

QUESTION 64

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Solve $y' = y/3$ with $y(0) = 10$.

A $10e^{-t/3}$

B $10 + t/3$

C $10e^{t/3}$

D $10e^{3t}$

QUESTION 65

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Solve $y' = -2y$ with $y(1) = 5$.

A $5e^{-2(x-1)}$

B $5e^{-2x}$

C $5 - 2(x - 1)$

D $5e^{2(x-1)}$

QUESTION 66

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Solve $y' = 2x/y$, $y(0) = 3$, with $y > 0$.

A $y = -\sqrt{(x^2 + 9)}$

B $y = \sqrt{(2x + 9)}$

C $y = x^2 + 9$

D $y = \sqrt{(x^2 + 9)}$

QUESTION 67

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Solve $y'=(1+t)y$ with $y(0)=2$.

A $2e^{t}$

B $2e^{t^2/2}$

C $2e^{t+t^2/2}$

D $2+t+t^2$

QUESTION 68

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $y'=ky$, $y(0)=4$, and $y(2)=16$, then $k=$

A $1/2$

B $\ln 2$

C $\ln 4$

D $2\ln 2$

QUESTION 69

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

An exponential model has half-life 6 hours and satisfies $A'=-kA$. The value of k is

A $\ln 2/6$

B $6/\ln 2$

C $-\ln 2/6$

D $\ln 6/2$

QUESTION 70

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Solve $y' = 5 - y$, $y(0) = 0$.

A $5(1+e^{-t})$

B $5(1-e^{-t})$

C $1-5e^{-t}$

D $5e^{-t}$

QUESTION 71

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $N' = 0.07N$ and $N(0) = 100$, $N(t)$ is

A $100e^{0.07}$

B $100e^{0.07t}$

C $100+0.07t$

D $100e^{-0.07t}$

QUESTION 72

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Under continuous growth at rate $k = 0.08$ per year, the doubling time is

A $0.08 \ln 2$

B $2/0.08$

C $\ln(0.08)/2$

D $\ln 2/0.08$

QUESTION 73

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $A' = -0.04A$, the half-life is

A $\ln(0.04)/2$

B $0.04/\ln 2$

C 25 hours

D $\ln 2/0.04$

QUESTION 74

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A population begins at 200 and grows continuously at 3% per year. After 10 years it is

A $200e^{3}$

B $200(1.03)^{10}$

C $200e^{0.3}$

D $200+30$

QUESTION 75

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

A substance follows $A' = -0.12A$. The fraction remaining after 5 time units is

A $e^{0.6}$

B $0.12e^{-5}$

C $1-0.6$

D $e^{-0.6}$

QUESTION 76

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $N'=0.05N$ and $N=400$ at an instant, then N' at that instant is

A 20
B 8
C 400
D 0.05

QUESTION 77

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

With continuous growth rate $k=0.06$, the time required for a quantity to triple is

A $3/0.06$
B $\ln 0.06/3$
C $\ln 3/0.06$
D $0.06 \ln 3$

QUESTION 78

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

A sample has half-life 12 hours. The fraction remaining after 18 hours is

A $1/8$
B $1/(2\sqrt{2})$
C $1/4$
D $1/2$

QUESTION 79

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $N(0)=50$ and $N(4)=100$ in an exponential model $N'=kN$, then $k=$

A $\ln 4/2$

B $4 \ln 2$

C $1/2$

D $\ln 2/4$

QUESTION 80

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $A'=-0.2A$, the time for A to fall to one quarter of its initial value is

A $\ln 2/0.2$

B $\ln 4/0.2$

C $0.2 \ln 4$

D $4/0.2$

QUESTION 81

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The solution of $T'=-0.2(T-20)$, $T(0)=80$, approaches which temperature as $t\to\infty$?

A $80°C$

B $60°C$

C $20°C$

D $0°C$

QUESTION 82

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $T' = -0.2(T - 20)$, when $T = 50$ the instantaneous rate T' is

A -10 degrees per time unit

B 20 degrees per time unit

C -6 degrees per time unit

D 6 degrees per time unit

QUESTION 83

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $P' = 0.3P(1 - P/800)$, the carrying capacity is

A 0

B 400

C 800

D 0.3

QUESTION 84

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The equilibrium values for $P' = 0.2P(1 - P/100)$ are

A $P = 100$ only

B $P = 20$ and $P = 100$

C $P = 0$ and $P = 100$

D $P = 0$ only

QUESTION 85

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For a standard logistic model $P'=rP(1-P/K)$, with $r,K>0$, the equilibrium $P=K$ is

A stable only if $P(0)>K$

B not an equilibrium

C stable

D unstable

QUESTION 86

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $P'=0.2P(1-P/100)$, the maximum growth rate occurs at $P=$

A 25

B 100

C 50

D 0

QUESTION 87

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

The general positive, non-equilibrium solution of $P'=rP(1-P/K)$, $r>0$, has the form

A $P=Ae^{rt}$

B $P=K(1+Ae^{-rt})$

C $P=K/(1-Ae^{rt})$

D $P=K/(1+Ae^{-rt})$

QUESTION 88

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $P'=0.5P(1-P/100)$, $P(0)=20$, the solution is

A $100/(1+4e^{-0.5t})$

B $100/(1-4e^{-0.5t})$

C $20e^{0.5t}$

D $100/(1+5e^{-0.5t})$

QUESTION 89

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

For $P'=0.4P(1-P/200)$, the growth rate when $P=100$ is

A 40

B 0

C 20

D 10

QUESTION 90

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

For $P'=0.5P(1-P/100)$, $P(0)=20$, the time when P first reaches 50 is

A $2\ln 4$

B $4\ln 2$

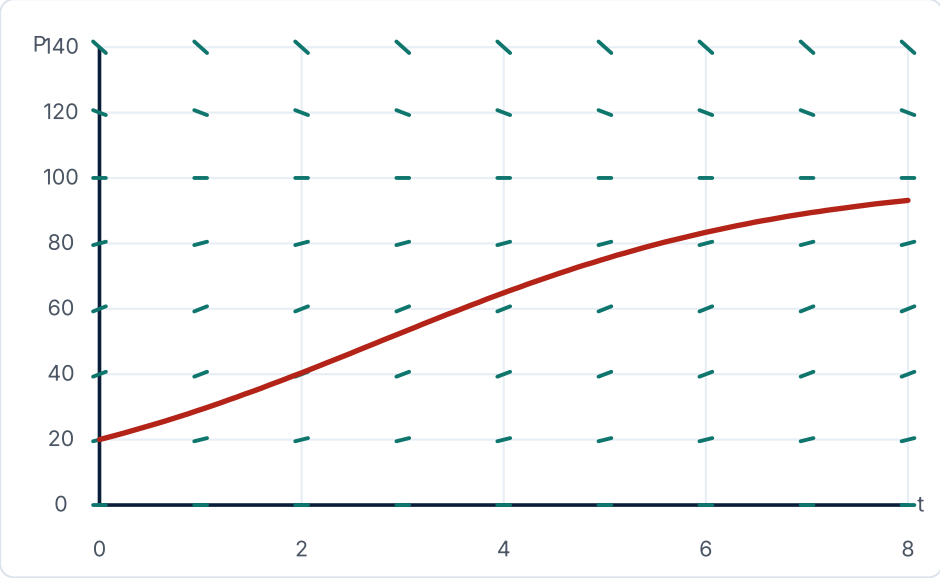
C $\ln 2$

D $\ln 4$

Question 91: Logistic Slope Field and Long-Term Behavior

★★★★★ Very Important | 9 points

A population P(t) satisfies $P' = 0.5P(1 - P/100)$. The slope field is shown below; the red curve is the solution with $P(0) = 20$.



Slope field for $P' = 0.5P(1 - P/100)$; red curve is the solution with $P(0) = 20$.

(a) [2 points] Find the equilibrium solutions of the differential equation.

(b) [2 points] Find the slope at the point (0,20), and interpret its sign.

(c) [2 points] Use the slope field to describe the long-term behavior of the solution starting at $P(0) = 20$.

(d) [3 points] Solve the initial-value problem and estimate $P(2)$.

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 92: Verifying and Applying a Solution

★★★★★ Very Important | 9 points

Consider the differential equation $y' = 2xy$ and the proposed family $y = Ce^{x^2}$.

(a) [2 points] Verify by differentiation that $y = Ce^{x^2}$ satisfies the differential equation.

(b) [2 points] Use $y(0) = 3$ to determine C and write the particular solution.

(c) [2 points] Evaluate $y(1)$.

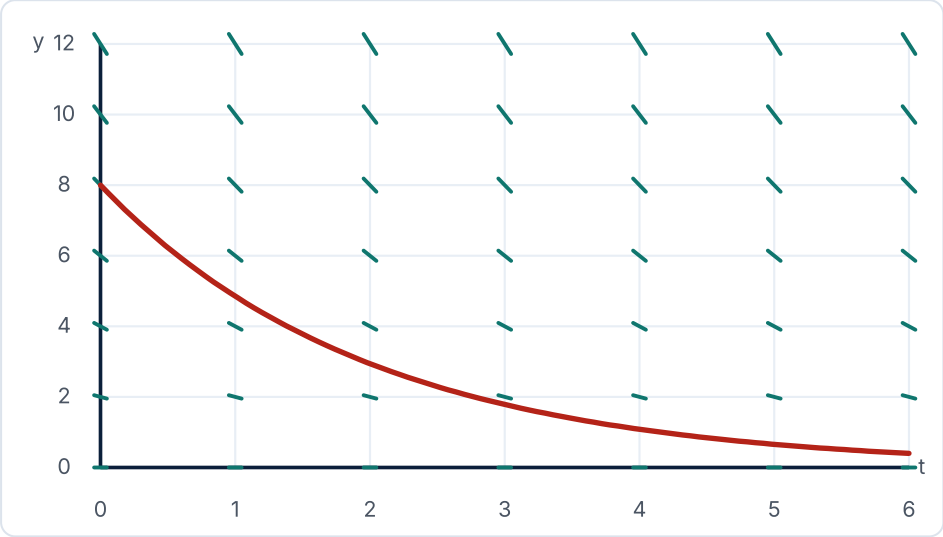
(d) [3 points] Show that $y = 0$ is also an equilibrium solution.

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 93: Reasoning from an Exponential-Decay Slope Field

★★★★★ Very Important | 9 points

The slope field for $y' = -0.5y$ is shown. A solution passes through $(0,8)$.



Slope field for $y' = -0.5y$; red curve passes through $(0,8)$.

(a) [2 points] Find the slope at $(0,8)$.

(b) [3 points] Separate variables and solve the initial-value problem.

(c) [2 points] Find $y(2)$.

(d) [2 points] State the equilibrium solution and the limit of $y(t)$ as $t \to \infty$.

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 94: Separation of Variables with an Initial Condition

★★★★★ Very Important | 9 points

A solution satisfies $y'=(x+1)/(y+1)$, $y(0)=1$, and $y>-1$.

(a) [2 points] Separate the variables.

(b) [3 points] Find the general implicit solution.

(c) [2 points] Use the initial condition to determine the constant.

(d) [2 points] Find $y(1)$.

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 95: Exponential Population Model from a Table

★★★★★ Very Important

9 points

A population follows $P' = kP$, where t is measured in hours. The table gives observed values.

t (hours)	0	2	4	6
P(t)	200	$200\sqrt{2}$	400	$400\sqrt{2}$

(a) [3 points] Use P(0) and P(4) to determine k.

(b) [2 points] Write the particular solution P(t).

(c) [2 points] Predict P(6).

(d) [2 points] Find $P'(4)$.

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 96: Radioactive Decay and Half-Life

★★★★★ Very Important | 9 points

A radioactive sample has initial mass 640 grams and follows $A' = -kA$. Its half-life is 8 hours.

(a) [2 points] Determine k.

(b) [2 points] Write A(t).

(c) [2 points] Find the mass after 24 hours.

(d) [3 points] How long does it take for the mass to reach 40 grams?

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 97: Determining a Logistic Model from Rates

★★★★★ Very Important | 9 points

A population P is modeled by $P' = rP(1 - P/K)$. Measurements are shown below.

P	0	50	100	150	200
P′	0	30	40	30	0

(a) [2 points] Use the table to identify the carrying capacity K.

(b) [2 points] Determine r.

(c) [3 points] Find the maximum growth rate and the population at which it occurs.

(d) [2 points] Classify the equilibria P=0 and P=200 as stable or unstable.

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 98: Solving a Logistic Initial-Value Problem

★★★★★ Very Important | 9 points

A population satisfies $P' = 0.4P(1 - P/200)$, $P(0) = 50$.

(a) [2 points] Find the equilibrium solutions.

(b) [3 points] Separate variables and obtain the logistic solution satisfying the initial condition.

(c) [2 points] Find the time when P first reaches 100.

(d) [2 points] State the long-term limit of P(t).

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 99: Logistic Model for Adoption

★★★★★ Very Important | 9 points

The number $A(t)$ of users adopting a service follows $A' = 0.6A(1 - A/1000)$, with $A(0) = 100$.

(a) [2 points] Identify the carrying capacity and both equilibrium solutions.

(b) [2 points] Write the particular logistic solution.

(c) [3 points] Find the time when the number of users reaches 500.

(d) [2 points] Find the initial adoption rate $A'(0)$.

Show your work. State the differential equation or theorem used and justify each conclusion.

Question 100: Newton's Cooling as a Differential Equation

★★★★★ Very Important | 9 points

A liquid cools in a room at 20°C. Its temperature T(t) satisfies $T' = -k(T-20)$, $k>0$. A table gives measurements.

t (minutes)	0	2	4	6
T(t) (°C)	100	60	40	30

(a) [2 points] State the equilibrium temperature and explain its meaning.

(b) [2 points] Use the data at t=0 and t=2 to determine k.

(c) [3 points] Write T(t) and verify that it agrees with the table at t=4.

(d) [2 points] Find $T'(0)$ and interpret its sign and units.

Show your work. State the differential equation or theorem used and justify each conclusion.

Questions 1–100 Complete

The next pages contain the MCQ answer key, worked solutions with distractor analysis, and FRQ model solutions with scoring rubrics.

> **Practice scoring:** 90 MCQs × 1 point plus 10 FRQs × 9 points = 180 raw points. This is not an official AP score conversion.

Multiple-Choice Answer Key

Displayed answer letters account for the deterministic option shuffle.

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
B ★★★★★	D ★★★★★	C ★★★	C ★★★★★	D ★★★★★	A ★★★★★	C ★★★★★	D ★★★	C ★★★	A ★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
C ★★★★★	C ★★★	B ★★★	B ★★★★★	B ★★★	D ★★★★★	B ★★★	A ★★★	B ★★★★★	D ★★★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
A ★★★	A ★★★	C ★★★	C ★★★	C ★★★	A ★★★★★	B ★★★	C ★★	C ★★★	D ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
A ★★★★★	D ★★★	D ★★★★★	C ★★★★★	A ★★★	B ★★★★★	C ★★★★★	A ★★★	B ★★★★★	D ★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
B ★★★	A ★★★	B ★★★	A ★★★★★	B ★★★	B ★★★★★	D ★★★	A ★★★	C ★★★★★	D ★★★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
B ★★★★★	D ★★★★★	D ★★★★★	A ★★★	D ★★★	D ★★★	D ★★★★★	A ★★★★★	B ★★★★★	B ★★★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
A ★★★★★	A ★★★★★	A ★★★★★	C ★★★	A ★★★	D ★★★★★	C ★★★★★	B ★★★★★	A ★★★★★	B ★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
B ★★★★★	D ★★★★★	D ★★★★★	C ★★★	D ★★★	A ★★★	C ★★★★★	B ★★★★★	D ★★★★★	B ★★★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
C ★★★★★	C ★★★	C ★★★★★	C ★★★★★	C ★★★★★	C ★★★★★	D ★★★★★	A ★★★★★	C ★★★	A ★★★★★

Question 01

★★★★★ Very Important

Correct answer: (B) $P'=0.04P$

STEP-BY-STEP SOLUTION

A constant per-capita growth rate of 0.04 means $P'/P=0.04$, so $P'=0.04P$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is a linear formula, not the stated rate law.

✗ **(C)** This is 400% per year, not 4%.

✗ **(D)** This describes a constant growth rate independent of population.

Question 02

★★★★★ Very Important

Correct answer: (D) $T'=-k(T-20)$, $k>0$

STEP-BY-STEP SOLUTION

The rate is proportional to the temperature difference from ambient, with sign opposite that difference: $T'=-k(T-20)$, $k>0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This would make the ambient equilibrium 0°C.

✗ **(B)** This gives the wrong ambient reference and sign.

✗ **(C)** This is a constant rate, not proportional to the temperature difference.

Question 03

★★★ Mandatory

Correct answer: (C) $P'=0.03P$

STEP-BY-STEP SOLUTION

Multiply the per-capita equation $P'/P=0.03$ by P to get $P'=0.03P$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is constant absolute growth, not constant per-capita growth.

✗ **(B)** This multiplies by the reciprocal incorrectly.

✗ **(D)** This is not a differential equation.

Question 04

★★★★★ Very Important

Correct answer: (C) $A'=12-0.2A$

STEP-BY-STEP SOLUTION

Net rate equals input rate minus removal rate: $A'=12-0.2A$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies the constant input by A.

✗ **(B)** Treats removal as an increase.

✗ **(D)** Reverses input and removal signs.

Question 05

★★★★★ Very Important

Correct answer: (D) 1000

STEP-BY-STEP SOLUTION

In logistic form $P'=rP(1-P/K)$, the carrying capacity is K. Here $K=1000$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the growth-rate parameter r.

✗ **(B)** This is an equilibrium, not the carrying capacity.

✗ **(C)** This is not the denominator in the logistic factor.

Question 06

★★★★★ Very Important

Correct answer: (A) $y=0$ and $y=5$

STEP-BY-STEP SOLUTION

Equilibria are constant solutions found by setting $y'=0$: $y(5-y)=0$ gives $y=0$ or $y=5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Misses $y=0$.

✗ **(C)** Solves the factors incorrectly.

✗ **(D)** Neither $y=1$ nor the pair shown solves $y(5-y)=0$.

Question 07

★★★★★ Very Important

Correct answer: (C) stable

STEP-BY-STEP SOLUTION

If $y<3$, then $y'>0$ and solutions rise toward 3. If $y>3$, then $y'<0$ and solutions fall toward 3. Thus $y=3$ is stable.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Solutions on both sides move toward 3.

✗ **(B)** Setting $3-y=0$ gives $y=3$.

✗ **(D)** The direction of motion on both sides establishes stability.

Question 08

★★★ Mandatory

Correct answer: (D) negative

STEP-BY-STEP SOLUTION

$1-60/50=-0.2$, so $P'=0.2(60)(-0.2)=-2.4$. The population is decreasing.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The right-hand side is defined at $P=60$.

✗ **(B)** This occurs at $P=0$ or $P=50$.

✗ **(C)** The logistic factor is negative above $K=50$.

Question 09

★★★ Mandatory

Correct answer: (C) per year

STEP-BY-STEP SOLUTION

P'/P has units (people/year)/people = 1/year. Thus k has units per year.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** That is a unit of P.

✗ **(B)** That is a unit of t.

✗ **(D)** That is the unit of P', not k.

Question 10

★★★ Mandatory

Correct answer: (A) 0 or 500

STEP-BY-STEP SOLUTION

Set the right-hand side equal to zero: $P=0$ or $1-P/500=0$, giving $P=500$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Misses the carrying-capacity equilibrium.

✗ **(C)** Misses the zero-population equilibrium.

✗ **(D)** $P=250$ gives positive growth, not zero growth.

Question 11

★★★★★ Very Important

Correct answer: (C) $y=Ce^{2x}$

STEP-BY-STEP SOLUTION

For $y=Ce^{2x}$, $y'=2Ce^{2x}=2y$, so it satisfies the differential equation.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Its derivative is $2Ce^{x}$, which is not generally $2(2Ce^{x})$.

✗ **(B)** Its derivative is $Ce^{x}=y$, not $2y$.

✗ **(D)** Its derivative is 2, not $2y$.

Question 12

★★★ Mandatory

Correct answer: (C) $y=x^2+C$

STEP-BY-STEP SOLUTION

Differentiating x^2+C gives $2x$, as required.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Its derivative is x.

✗ **(B)** Its derivative is $4x$.

✗ **(D)** Its derivative is 1.

Question 13

★★★ Mandatory

Correct answer: (B) $y=5e^{-t}$

STEP-BY-STEP SOLUTION

The derivative of $5e^{-t}$ is $-5e^{-t}$, which equals $-y$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Its derivative is -1, not $-y$.

✗ **(C)** Its derivative is y, not $-y$.

✗ **(D)** Its derivative is $-e^{-t}$, which does not equal $-(e^{-t}+5)$.

Question 14

★★★★★ Very Important

Correct answer: (B) $y=2+Ce^{-3x}$

STEP-BY-STEP SOLUTION

For $y=2+Ce^{-3x}$, $y'=-3Ce^{-3x}=-3(y-2)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This does not satisfy the equation for general x.

✗ **(C)** Differentiation gives $+3(y-2)$, the wrong sign.

✗ **(D)** This omits the equilibrium shift 2.

Question 15

★★★ Mandatory

Correct answer: (B) $y=0$

STEP-BY-STEP SOLUTION

A constant solution requires $y'=0$ and $y(5-y)=0$. $y=0$ is one such equilibrium solution ($y=5$ is the other).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The right side is 4, not zero.

✗ **(C)** The right side is nonzero.

✗ **(D)** The right side is nonzero.

Question 16

★★★★★ Very Important

Correct answer: (D) $y'=y(4-y)$

STEP-BY-STEP SOLUTION

Let $D=1+3e^{-4t}$. Then $y'=48e^{-4t}/D^2$. Also $y(4-y)=(4/D)(12e^{-4t}/D)=48e^{-4t}/D^2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This has logistic rate 1, not rate 4.

✗ **(B)** The solution is not a pure exponential.

✗ **(C)** The derivative is not $-4y$.

Question 17

★★★ Mandatory

Correct answer: (B) $y=\ln x+C$

STEP-BY-STEP SOLUTION

The derivative of $\ln x+C$ is $1/x$ for $x>0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Its derivative is $-1/x^2$.

✗ **(C)** Its derivative is 1.

✗ **(D)** Its derivative is e^x.

Question 18

★★★ Mandatory

Correct answer: (A) $y=\sin x+C$

STEP-BY-STEP SOLUTION

The derivative of $\sin x+C$ is $\cos x$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Its derivative is $-\cos x$.

✗ **(C)** Its derivative is $\sin x$.

✗ **(D)** Its derivative is $-\sin x$.

Question 19

★★★★★ Very Important

Correct answer: (B) $y=Ce^{x^2}$

STEP-BY-STEP SOLUTION

For $y=Ce^{x^2}$, $y'=2xCe^{x^2}=2xy$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Its derivative is 2y, not 2xy.

✗ **(C)** Its derivative is y, not 2xy.

✗ **(D)** Its derivative is 2Cx, not 2xy in general.

Question 20

★★★★★ Very Important

Correct answer: (D) $y=3+Ce^{-t}$

STEP-BY-STEP SOLUTION

For $y=3+Ce^{-t}$, $y'=-Ce^{-t}$, and $3-y=-Ce^{-t}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Its derivative is 3, not 3−y.

✗ **(B)** This does not include the equilibrium shift 3.

✗ **(C)** The sign of the exponential gives $y'=y-3$, not 3−y.

Question 21

★★★ Mandatory

Correct answer: (A) 1

STEP-BY-STEP SOLUTION

Substitute $x=2$ and $y=-1$: $y'=2-1=1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds absolute values instead of substituting.

✗ **(C)** Arithmetic/sign error.

✗ **(D)** This omits the x contribution.

Question 22

★★★ Mandatory

Correct answer: (A) −1

STEP-BY-STEP SOLUTION

$y' = 2(1) - 3 = -1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds 2 and 3.

✗ **(C)** Uses only 3.

✗ **(D)** Drops the negative y term.

Question 23

★★★ Mandatory

Correct answer: (C) 4

STEP-BY-STEP SOLUTION

At $y=2$, $y' = 2(4-2) = 4$. This equation is autonomous, so the slope is the same for every x on that row.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The slope is zero only at $y=0$ or 4.

✗ **(B)** This omits a factor of 2.

✗ **(D)** The right-hand side is positive at $y=2$.

Question 24

★★★ Mandatory

Correct answer: (C) x=3

STEP-BY-STEP SOLUTION

A horizontal segment means $y'=0$. Solve $3-x=0$ to get $x=3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** At $x=0$ the slope is 3.

✗ **(B)** The equation does not depend on y.

✗ **(D)** At $y=0$ the slope still depends on x.

Question 25

★★★ Mandatory

Correct answer: (C) y=2

STEP-BY-STEP SOLUTION

Set $2-y=0$. The zero-slope isocline is the horizontal row $y=2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This relation is not the zero-slope condition.

✗ **(B)** The equation has no x term.

✗ **(D)** At $y=0$ the slope is 2.

Question 26

★★★★★ Very Important

Correct answer: (A) x=1 or y=−2

STEP-BY-STEP SOLUTION

A product is zero if either factor is zero: $x-1=0$ or $y+2=0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is unrelated to the factors.

✗ **(C)** Every point on either zero-slope line has slope zero, not just their intersection.

✗ **(D)** Both zero conditions have the wrong signs.

Question 27

★★★ Mandatory

Correct answer: (B) x=−3 and x=3

STEP-BY-STEP SOLUTION

Set $x^2-9=0$, so $x=\pm 3$. Since the equation depends only on x, these are vertical lines in the slope field.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses both roots.

✗ **(C)** These would be horizontal lines, but the slope equation has no y.

✗ **(D)** Misses $x=-3$.

Question 28

★★ Important

Correct answer: (C) 0

STEP-BY-STEP SOLUTION

Substitute $x=y=2$: $y'=2-2=0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds x and y.

✗ **(B)** Arithmetic error.

✗ **(D)** Uses x only.

Question 29

★★★ Mandatory

Correct answer: (C) Slopes are positive at every point

STEP-BY-STEP SOLUTION

Since $1+y^2 \geq 1>0$ for every real y, every slope is positive.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The slope remains positive even when $y<0$.

✗ **(B)** At $y=0$ the slope is 1.

✗ **(D)** The slope depends on y.

Question 30

★★★ Mandatory

Correct answer: (D) 0

STEP-BY-STEP SOLUTION

Substitute $y=0$: $y'=-0=0$. Thus $y=0$ is an equilibrium solution.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $y=-1$.

✗ **(B)** Uses $y=1$.

✗ **(C)** The equation is autonomous; at $y=0$ the slope is fixed.

Question 31

★★★★★ Very Important

Correct answer: (A) increasing

STEP-BY-STEP SOLUTION

At $y=10$, $y'=0.5(10)(1-10/20)=2.5>0$, so the solution is increasing.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** The equilibrium values are 0 and 20.

✗ **(C)** The equation is defined at $y=10$.

✗ **(D)** The derivative is positive.

Question 32

★★★ Mandatory

Correct answer: (D) decreasing

STEP-BY-STEP SOLUTION

Since $y>20$, $1-y/20<0$; with y positive, $y'<0$. The solution decreases.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** 25 is not an equilibrium.

✗ **(B)** The derivative is negative above carrying capacity.

✗ **(C)** The right-hand side is defined.

Question 33

★★★★★ Very Important

Correct answer: (D) y=0 and y=20

STEP-BY-STEP SOLUTION

Set $y'=0$. The factors give $y=0$ or $1-y/20=0$, hence $y=20$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses the zero solution.

✗ **(B)** Misses the carrying-capacity equilibrium.

✗ **(C)** $y=10$ is not an equilibrium.

Question 34

★★★★★ Very Important

Correct answer: (C) stable

STEP-BY-STEP SOLUTION

Below 4 the derivative is positive; above 4 it is negative. Solutions move toward y=4 from both sides, so it is stable.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Substitution gives $y'=0$ at $y=4$.

✗ **(B)** The constant solution $y=4$ works for every x.

✗ **(D)** Both sides move toward 4, not away.

Question 35

★★★ Mandatory

Correct answer: (A) unstable

STEP-BY-STEP SOLUTION

Below 4 the slope is negative, so solutions move lower; above 4 the slope is positive, so they move higher. Both directions move away from 4.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Stability is about solution behavior, not a local maximum of a function.

✗ **(C)** Substitution gives zero at $y=4$.

✗ **(D)** The field points away from the equilibrium.

Question 36

★★★★★ Very Important

Correct answer: (B) equal to 0

STEP-BY-STEP SOLUTION

$y=0$ is an equilibrium solution. The solution starting exactly at 0 stays at 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The derivative is zero at $y=0$.

✗ **(C)** The initial condition is not $y=5$.

✗ **(D)** The differential equation gives $y'=0$ at $y=0$.

Question 37

★★★★★ Very Important

Correct answer: (C) increases toward 5

STEP-BY-STEP SOLUTION

For 00. As y approaches 5, the slope approaches 0; the solution increases toward carrying capacity 5.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The derivative is positive on this interval.

✗ **(B)** The slope changes sign at y=5 and solutions approach that equilibrium.

✗ **(D)** It is not an equilibrium unless y is 0 or 5.

Question 38

★★★ Mandatory

Correct answer: (A) decreases toward 5

STEP-BY-STEP SOLUTION

At y=7, $y'=7(-2)<0$, so the solution decreases toward the stable equilibrium y=5.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** 7 is not an equilibrium.

✗ **(C)** The stable equilibrium attracts solutions from above.

✗ **(D)** The slope is negative above 5.

Question 39

★★★★★ Very Important

Correct answer: (B) decreases toward 0

STEP-BY-STEP SOLUTION

The solution is $y=y(0)e^{-2x}$, which stays positive and decreases toward 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The exponential solution remains positive for finite x.

✗ **(C)** Only y=0 is constant.

✗ **(D)** The derivative is negative when y>0.

Question 40

★★★ Mandatory

Correct answer: (D) 4

STEP-BY-STEP SOLUTION

$y'=(-2)^2=4$, so the slope is positive.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses y rather than y^2.

✗ **(B)** The derivative is not zero.

✗ **(C)** Sign error when squaring.

Question 41

★★★ Mandatory

Correct answer: (B) 3

STEP-BY-STEP SOLUTION

Net change is $\int_0^2 x\,dx=2$. Thus $y(2)=1+2=3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the starting value.

✗ **(C)** Uses 2^2 rather than the integral.

✗ **(D)** Adds $1+\int_0^2 2x\,dx$.

Question 42

★★★ Mandatory

Correct answer: (A) 11

STEP-BY-STEP SOLUTION

$y(3)=y(1)+\int_1^3 2x\,dx=3+[x^2]_1^3=3+8=11$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This omits the initial value or miscomputes the change.

✗ **(C)** Uses the upper endpoint only.

✗ **(D)** Subtracts the initial value.

Question 43

★★★ Mandatory

Correct answer: (B) 1

STEP-BY-STEP SOLUTION

$y(\pi)=1+\int_0^{\pi} \cos x\,dx=1+\sin\pi-\sin 0=1.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses interval length with solution value.

✗ **(C)** Adds the absolute area instead of signed change.

✗ **(D)** The net change is zero, but the initial value is 1.

Question 44

★★★★★ Very Important

Correct answer: (A) $y=-1$

STEP-BY-STEP SOLUTION

For $y<-1$ the product is positive, so solutions rise toward -1. For -1

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Solutions move away from 2 on both sides.

✗ **(C)** The two equilibria have different stability.

✗ **(D)** One equilibrium is stable.

Question 45

★★★ Mandatory

Correct answer: (B) $y=3x^2+C$

STEP-BY-STEP SOLUTION

Integrate: $y=\int 6x\,dx=3x^2+C.$

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiating this gives $12x$.

✗ **(C)** Its derivative is 3.

✗ **(D)** Its derivative is $6x^5$.

Question 46

★★★★★ Very Important

Correct answer: (B) $y=Ce^{x^2}$

STEP-BY-STEP SOLUTION

Separate: $dy/y=2x\,dx$. Integrate $\ln|y|=x^2+C$, so $y=Ce^{x^2}$ (with C any constant, including 0).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The exponent should be x^2, not 2x.

✗ **(C)** Its derivative is y, not 2xy.

✗ **(D)** This does not solve the equation for arbitrary C.

Question 47

★★★ Mandatory

Correct answer: (D) $y=Cx$

STEP-BY-STEP SOLUTION

Separate $dy/y=dx/x$. Integrating gives $\ln|y|=\ln|x|+C$, hence $y=Cx$ on each connected interval not crossing $x=0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This would give $y'=2Cx$.

✗ **(B)** This is the reciprocal dependence.

✗ **(C)** This is not the separated solution.

Question 48

★★★ Mandatory

Correct answer: (A) $y^2=x^2+C$

STEP-BY-STEP SOLUTION

Separate $y\,dy=x\,dx$. Integrate $y^2/2=x^2/2+C$; multiplying by 2 and absorbing constants gives $y^2=x^2+C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** The x integral is $x^2/2$, not $x/2$.

✗ **(C)** The powers and variables are wrong.

✗ **(D)** The sign of the x integral is positive.

Question 49

★★★★★ Very Important

Correct answer: (C) $y^2/2+y=x^2/2+x+C$

STEP-BY-STEP SOLUTION

Separate: $(y+1)dy=(x+1)dx$. Integrate each side to obtain $y^2/2+y=x^2/2+x+C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Does not follow from integrating the separated equation.

✗ **(B)** This is missing integration of $y+1$ and the correct relation.

✗ **(D)** The right-hand integral is $x^2/2+x$, not x^2+x.

Question 50

★★★★★ Very Important

Correct answer: (D) $e^y=e^x+C$

STEP-BY-STEP SOLUTION

Rewrite $y'=e^xe^{-y}$. Multiply by e^y: $e^ydy=e^xdx$. Integrating gives $e^y=e^x+C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The left side integrates to e^y, not y.

✗ **(B)** The separation uses e^ydy, not $e^{-y}dy$.

✗ **(C)** This is not the result of the separated integrals.

Question 51

★★★★★ Very Important

Correct answer: (B) $y^3=x^3+C$

STEP-BY-STEP SOLUTION

Separate $y^2dy=x^2dx$. Integrate: $y^3/3=x^3/3+C$. Multiply by 3 and absorb the constant to get $y^3=x^3+C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrating x^2 gives $x^3/3$, not x^2.

✗ **(C)** Integrating y^2 gives $y^3/3$, not y^2.

✗ **(D)** The left side must be integrated with respect to y.

Question 52

★★★★★ Very Important

Correct answer: (D) arctan y=x+C

STEP-BY-STEP SOLUTION

Separate $dy/(1+y^2)=dx$. Integrating gives arctan y=x+C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This reverses the variables incorrectly.

✗ **(B)** The integral of $1/(1+y^2)$ is arctan y.

✗ **(C)** The solution is implicit after separation; the stated form is not the general antiderivative.

Question 53

★★★★★ Very Important

Correct answer: (D) y=C e^(arctan x)

STEP-BY-STEP SOLUTION

Separate $dy/y=dx/(1+x^2)$. Integrate $\ln|y|=\arctan x+C$, so y=C e^(arctan x).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This omits the logarithmic/exponential relationship.

✗ **(B)** Differentiation does not produce $y/(1+x^2)$.

✗ **(C)** The separated integral on the right is arctan x, not x.

Question 54

★★★ Mandatory

Correct answer: (A) $y^2=2\sin x+C$

STEP-BY-STEP SOLUTION

Separate $y\,dy=\cos x\,dx$. Integrate $y^2/2=\sin x+C$, then multiply by 2 and absorb the constant.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Misses the factor 2 after multiplying through.

✗ **(C)** The antiderivative of cos x is positive sin x.

✗ **(D)** Integrates with respect to the wrong variable.

Question 55

★★★ Mandatory

Correct answer: (D) (2/3)y^(3/2)=x²/2+x+C

STEP-BY-STEP SOLUTION

Separate √y dy=(x+1)dx. Integrate: (2/3)y^(3/2)=x²/2+x+C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The integral of √y is (2/3)y^(3/2), not 2√y.

✗ **(B)** Misses the factor 2/3 and the 1/2 on x².

✗ **(C)** Integrates √y with the wrong power.

Question 56

★★★ Mandatory

Correct answer: (D) y=C(x+1)

STEP-BY-STEP SOLUTION

Separate dy/y=dx/(x+1). Then ln|y|=ln(x+1)+C, so y=C(x+1).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds an extra power.

✗ **(B)** Exponentiates the wrong integral.

✗ **(C)** This is the reciprocal dependence.

Question 57

★★★★★ Very Important

Correct answer: (D) y=−1/(x+C)

STEP-BY-STEP SOLUTION

Separate $y^{-2}dy=dx$. Integrate −1/y=x+C, so y=−1/(x+C). The zero equilibrium is excluded by the stem's nonzero condition.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The sign is opposite to the integrated result.

✗ **(B)** This solves y′=y, not y′=y².

✗ **(C)** This is not the result after solving for y.

Question 58

★★★★★ Very Important

Correct answer: (A) $4/(1+3e^{-x})$

STEP-BY-STEP SOLUTION

The logistic form is $y=4/(1+Ae^{-x})$. Using $y(0)=1$ gives $1=4/(1+A)$, so $A=3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** At $x=0$ the denominator is -2, giving the wrong initial value.

✗ **(C)** Its carrying capacity is 1, not 4.

✗ **(D)** At $x=0$ this gives 2, not 1.

Question 59

★★★★★ Very Important

Correct answer: (B) $y^2=x^2+4$

STEP-BY-STEP SOLUTION

Separate $y\,dy=x\,dx$ and integrate $y^2/2=x^2/2+C$. At $x=0, y=2$ gives $C=2$, hence $y^2=x^2+4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The integral of x is $x^2/2$, not x.

✗ **(C)** The initial condition gives $y^2(0)=4$, so the constant is 4.

✗ **(D)** This is not the integrated relation.

Question 60

★★★★★ Very Important

Correct answer: (B) $y+y^2/2=x^2+3/2$

STEP-BY-STEP SOLUTION

Separate $(1+y)dy=2x\,dx$. Integrate $y+y^2/2=x^2+C$. At $(0,1)$, $C=1+1/2=3/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The coefficient on y^2 should be $1/2$.

✗ **(C)** The initial value at $x=0$ gives $3/2$, not 1.

✗ **(D)** Omits the y term from integrating $1+y$.

Question 61

★★★★★ Very Important

Correct answer: (A) $3e^{x^2}$

STEP-BY-STEP SOLUTION

The general solution is Ce^{x^2}. Applying $y(0)=3$ gives $C=3$, so $y=3e^{x^2}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds the initial value rather than setting the constant.

✗ **(C)** Uses the wrong exponent.

✗ **(D)** This is not an exponential solution.

Question 62

★★★★★ Very Important

Correct answer: (A) $80e^{-0.5t}$

STEP-BY-STEP SOLUTION

For $y'=ky$, $y=Ce^{kt}$. Here $C=80$ and $k=-0.5$, so $y=80e^{-0.5t}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** A constant additive decay rate is a different model.

✗ **(C)** This grows instead of decaying.

✗ **(D)** This has the wrong initial value and decay rate.

Question 63

★★★★★ Very Important

Correct answer: (A) $4-3e^{-t}$

STEP-BY-STEP SOLUTION

Let $z=y-4$. Then $z'=-z$, so $y=4+Ce^{-t}$. The initial value gives $C=-3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** At $t=0$ this equals 5.

✗ **(C)** At $t=0$ this equals 7.

✗ **(D)** At $t=0$ this equals 3 and has equilibrium 0.

Question 64

★★★ Mandatory

Correct answer: (C) $10e^{t/3}$

STEP-BY-STEP SOLUTION

The equation is $y'=(1/3)y$, so $y=Ce^{t/3}$. The initial condition gives $C=10$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the wrong sign.

✗ **(B)** This is not an exponential model.

✗ **(D)** Uses rate 3 rather than 1/3.

Question 65

★★★ Mandatory

Correct answer: (A) $5e^{-2(x-1)}$

STEP-BY-STEP SOLUTION

The solution anchored at $x=1$ is $y=5e^{-2}(x-1)$, which gives $y(1)=5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** At $x=1$ this is $5e^{-2}$, not 5.

✗ **(C)** This is linear, not exponential.

✗ **(D)** This grows and does not satisfy $y'=-2y$.

Question 66

★★★★★ Very Important

Correct answer: (D) $y=\sqrt{(x^2+9)}$

STEP-BY-STEP SOLUTION

Separate $y\,dy=2x\,dx$. Then $y^2/2=x^2+C$. At $x=0,y=3$ gives $C=9/2$, so $y^2=x^2+9$. Since $y>0$, take the positive square root.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Violates $y>0$ and the initial condition.

✗ **(B)** Integrating $2x$ gives x^2, not $2x$.

✗ **(C)** The solution is not $y^2=x^2+9$ as stated.

Question 67

★★★★★ Very Important

Correct answer: (C) $2e^{t+t^2/2}$

STEP-BY-STEP SOLUTION

Separate $dy/y=(1+t)dt$. Integrate $\ln|y|=t+t^2/2+C$. The initial condition gives $y=2e^{t+t^2/2}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the $t^2/2$ term.

✗ **(B)** Omits the t term.

✗ **(D)** This is not the exponential solution.

Question 68

★★★★★ Very Important

Correct answer: (B) ln 2

STEP-BY-STEP SOLUTION

$y=4e^{kt}$. At $t=2$, $16=4e^{2k}$, so $e^{2k}=4$ and $2k=\ln 4=2\ln 2$. Therefore $k=\ln 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This does not satisfy $e^{2k}=4$.

✗ **(C)** This is 2k, not k.

✗ **(D)** This equals ln4, which is 2k.

Question 69

★★★★★ Very Important

Correct answer: (A) ln2/6

STEP-BY-STEP SOLUTION

$A(t)=A_0e^{-kt}$. Half-life means $e^{-6k}=1/2$, so $6k=\ln 2$ and $k=\ln 2/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is the reciprocal rate.

✗ **(C)** k is positive for decay in $A'=-kA$.

✗ **(D)** Does not follow from the half-life equation.

Question 70

★★★ Mandatory

Correct answer: (B) $5(1-e^{-t})$

STEP-BY-STEP SOLUTION

The general solution is $y=5+Ce^{-t}$. The initial value gives $C=-5$, so $y=5(1-e^{-t})$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** At $t=0$ this equals 10.

✗ **(C)** At $t=0$ this equals -4.

✗ **(D)** This tends to 0 rather than the equilibrium 5.

Question 71

★★★★★ Very Important

Correct answer: (B) $100e^{0.07t}$

STEP-BY-STEP SOLUTION

The solution of $N'=kN$ with $N(0)=N_0$ is $N=N_0e^{kt}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is constant and does not depend on t.

✗ **(C)** This is linear growth, not proportional growth.

✗ **(D)** This models decay, not positive growth.

Question 72

★★★★★ Very Important

Correct answer: (D) $\ln 2/0.08$

STEP-BY-STEP SOLUTION

For doubling, $e^{kt}=2$. Thus $t=\ln 2/k=\ln 2/0.08$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This has units/rate arranged incorrectly.

✗ **(B)** The exponential doubling condition involves $\ln 2$.

✗ **(C)** This is not obtained from $e^{kt}=2$.

Question 73

★★★★★ Very Important

Correct answer: (D) ln2/0.04

STEP-BY-STEP SOLUTION

At half-life t, $e^{-0.04t}=1/2$. Hence $t=\ln 2/0.04$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Does not solve the half-life equation.

✗ **(B)** Inverts the rate incorrectly.

✗ **(C)** The value 25 is 1/0.04 but misses the ln2 factor.

Question 74

★★★ Mandatory

Correct answer: (C) $200e^{0.3}$

STEP-BY-STEP SOLUTION

The continuous exponential model is $P(t)=200e^{0.03t}$. At $t=10$, $P=200e^{0.3}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 3 instead of 0.3 in the exponent.

✗ **(B)** This is discrete annual growth, not the stated continuous model.

✗ **(D)** Adds a constant amount each year.

Question 75

★★★ Mandatory

Correct answer: (D) $e^{-0.6}$

STEP-BY-STEP SOLUTION

$A(t)/A_0=e^{-0.12t}$. At $t=5$, the fraction is $e^{-0.6}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This represents growth.

✗ **(B)** Includes an unnecessary factor and wrong exponent.

✗ **(C)** A constant-rate linear fraction is not the model.

Question 76

★★★ Mandatory

Correct answer: (A) 20

STEP-BY-STEP SOLUTION

Substitute $N=400$: $N'=0.05(400)=20$ units per time.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Divides rather than multiplies.

✗ **(C)** Confuses the population with its rate.

✗ **(D)** Confuses the growth parameter with the current rate.

Question 77

★★★★★ Very Important

Correct answer: (C) ln3/0.06

STEP-BY-STEP SOLUTION

$e^{kt}=3$, so $t=\ln 3/k=\ln 3/0.06$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 3 instead of ln3.

✗ **(B)** Does not solve the tripling equation.

✗ **(D)** Multiplies instead of dividing by k.

Question 78

★★★★★ Very Important

Correct answer: (B) $1/(2\sqrt{2})$

STEP-BY-STEP SOLUTION

Eighteen hours is 1.5 half-lives, so the fraction is $(1/2)^{1.5}=1/(2\sqrt{2})$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the fraction after three half-lives.

✗ **(C)** This is the fraction after two half-lives.

✗ **(D)** This is the fraction after one half-life.

Question 79

★★★★★ Very Important

Correct answer: (D) ln2/4

STEP-BY-STEP SOLUTION

$100=50e^{4k}$, so $e^{4k}=2$ and $k=\ln 2/4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This equals $\ln 4/2=\ln 2$, which is 4 times too large.

✗ **(B)** This is the wrong arrangement.

✗ **(C)** This is not obtained from the exponential equation.

Question 80

★★★★★ Very Important

Correct answer: (B) ln4/0.2

STEP-BY-STEP SOLUTION

$A/A_0=e^{-0.2t}$. Set $e^{-0.2t}=1/4$; then $0.2t=\ln 4$, so $t=\ln 4/0.2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the half-life, not the quarter-life.

✗ **(C)** Multiplies by the rate instead of dividing.

✗ **(D)** Uses 4 rather than ln4.

Question 81

★★★★★ Very Important

Correct answer: (C) 20°C

STEP-BY-STEP SOLUTION

The equilibrium occurs when $T'=0$, so $T=20$. The solution is $T=20+60e^{-0.2t}$, which tends to 20°C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** 80 is the initial temperature.

✗ **(B)** 60 is the initial temperature difference, not the limiting temperature.

✗ **(D)** The model's equilibrium is not zero.

Question 82

★★★ Mandatory

Correct answer: (C) −6 degrees per time unit

STEP-BY-STEP SOLUTION

$T'=-0.2(50-20)=-0.2(30)=-6$. The negative sign indicates cooling.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 50 rather than the temperature difference 30.

✗ **(B)** Confuses the ambient temperature with the rate.

✗ **(D)** Drops the negative sign.

Question 83

★★★★★ Very Important

Correct answer: (C) 800

STEP-BY-STEP SOLUTION

Compare with $P'=rP(1-P/K)$; the denominator K is the carrying capacity.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Zero is an equilibrium, not the carrying capacity.

✗ **(B)** Half the carrying capacity.

✗ **(D)** This is the intrinsic growth-rate parameter.

Question 84

★★★★★ Very Important

Correct answer: (C) P=0 and P=100

STEP-BY-STEP SOLUTION

Set $P'=0$. The factors give $P=0$ or $1-P/100=0$, so $P=100$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses the zero-population equilibrium.

✗ **(B)** 20 is not an equilibrium.

✗ **(D)** Misses the carrying-capacity equilibrium.

Question 85

★★★★★ Very Important

Correct answer: (C) stable

STEP-BY-STEP SOLUTION

For 00; for P>K, P′<0. Solutions on both sides move toward K, so K is stable.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Stability holds from both sides.

✗ **(B)** Substitution gives P′=0 at P=K.

✗ **(D)** Solutions move toward K from both sides.

Question 86

★★★★★ Very Important

Correct answer: (C) 50

STEP-BY-STEP SOLUTION

The logistic rate rP(1−P/K) is a downward-opening quadratic in P, maximized halfway to carrying capacity: P=K/2=50.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is not K/2.

✗ **(B)** Growth is zero at carrying capacity.

✗ **(D)** Growth is zero at P=0.

Question 87

★★★★★ Very Important

Correct answer: (D) $P=K/(1+Ae^{-rt})$

STEP-BY-STEP SOLUTION

Separating variables and integrating gives the logistic family $P(t)=K/(1+Ae^{-rt})$, with A determined by the initial condition.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is an exponential model, not logistic.

✗ **(B)** This does not approach K in the standard way.

✗ **(C)** The exponent/sign form is not the standard general logistic solution.

Question 88

★★★★★ Very Important

Correct answer: (A) $100/(1+4e^{-0.5t})$

STEP-BY-STEP SOLUTION

The logistic form is $100/(1+Ae^{-0.5t})$. At $t=0$, $20=100/(1+A)$, so $A=4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** At $t=0$ this gives 20, but it does not satisfy the standard logistic growth form for all t.

✗ **(C)** This ignores the carrying capacity.

✗ **(D)** At $t=0$ this gives $100/6$, not 20.

Question 89

★★★ Mandatory

Correct answer: (C) 20

STEP-BY-STEP SOLUTION

$P'=0.4(100)(1-100/200)=40(1/2)=20$ population units per time unit.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor $1-P/K$.

✗ **(B)** The growth is maximal at half of K, not zero.

✗ **(D)** Half the correct result.

Question 90

★★★★★ Very Important

Correct answer: (A) $2\ln 4$

STEP-BY-STEP SOLUTION

$P(t)=100/(1+4e^{-0.5t})$. Set $P=50$: $2=1+4e^{-0.5t}$, so $e^{-0.5t}=1/4$. Thus $t=2\ln 4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses the reciprocal time factor.

✗ **(C)** Uses the wrong initial-condition constant.

✗ **(D)** Misses the factor 2 from the exponent coefficient 0.5.

Question 91: Logistic Slope Field and Long-Term Behavior

★★★★★ Very Important

MODEL SOLUTION

(a) Set $P'=0$: $P=0$ or $1-P/100=0$. Equilibria are **P=0 and P=100**.

(b) At $P=20$, $P'=0.5(20)(1-0.2)=10(0.8)=$**8 population units per time unit**. The positive sign means P is increasing at that instant.

(c) Starting below 100 and above 0, the solution increases and levels off toward the stable equilibrium **P=100**.

(d) The logistic solution is $P=100/(1+Ae^{-0.5t})$. $P(0)=20$ gives $A=4$. Thus $P(t)=100/(1+4e^{-0.5t})$. At $t=2$, $P(2)=100/(1+4/e)\approx$**40.5**.

SCORING RUBRIC

Criterion	Points
Both equilibria P=0 and P=100	2
Slope 8 and correct interpretation	2
Increasing toward 100 from P(0)=20	2
Correct logistic solution and A=4	2
Correct estimate P(2)≈40.5	1
Total	9

Question 92: Verifying and Applying a Solution

★★★★★ Very Important

MODEL SOLUTION

(a) $y' = 2xCe^{x^2}$. Since $y = Ce^{x^2}$, this is $2xy$, so the family satisfies $y' = 2xy$.
(b) $y(0) = C = 3$, so $\mathbf{y = 3e^{x^2}}$.
(c) $y(1) = \mathbf{3e}$.
(d) If $y = 0$ for every x, then $y' = 0$ and the right side $2x(0) = 0$. Therefore $\mathbf{y = 0}$ is also a solution.

SCORING RUBRIC

Criterion	Points
Correct derivative using chain rule	2
Uses initial condition to get $C = 3$	2
Correct value $3e$	2
Checks both sides for $y = 0$	3
Total	9

Question 93: Reasoning from an Exponential-Decay Slope Field

★★★★★ Very Important

MODEL SOLUTION

(a) $y' = -0.5(8) = \mathbf{-4}$.

(b) $dy/y = -0.5dt$. Integrate: $\ln|y| = -0.5t + C$, so $y = Ce^{-0.5t}$. From $y(0) = 8$, $C = 8$. Thus $\mathbf{y = 8e^{-0.5t}}$.

(c) $y(2) = 8e^{-1} = \mathbf{8/e}$.

(d) The equilibrium is $\mathbf{y = 0}$; the given positive solution decreases toward 0, so $\lim y(t) = \mathbf{0}$.

SCORING RUBRIC

Criterion	Points
Slope -4	2
Correct separation/integration	2
Applies initial condition to obtain $8e^{-0.5t}$	1
$y(2) = 8/e$	2
Equilibrium and limit 0	2
Total	9

Question 94: Separation of Variables with an Initial Condition

★★★★★ Very Important

MODEL SOLUTION

(a) Multiply both sides by $(y+1)dx$: $\mathbf{(y+1)dy=(x+1)dx}$.

(b) Integrating gives $y+y^2/2=x+x^2/2+C$.

(c) At (0,1), $C=1+1/2=\mathbf{3/2}$. The particular relation is $y+y^2/2=x+x^2/2+3/2$.

(d) At $x=1$, $y+y^2/2=3$. Multiply by 2: $y^2+2y-6=0$. Thus $y=-1\pm\sqrt{7}$; $y>-1$ selects $\mathbf{y(1)=\sqrt{7}-1}$.

SCORING RUBRIC

Criterion	Points
Correct separation	2
Integrates both sides correctly	2
Includes integration constant	1
Finds $C=3/2$	2
Solves and selects $\sqrt{7}-1$ using $y>-1$	2
Total	9

Question 95: Exponential Population Model from a Table

★★★★★ Very Important

MODEL SOLUTION

(a) $P(t)=P_0e^{kt}$. Using $400=200e^{4k}$ gives $e^{4k}=2$, so **$k=\ln 2/4$** per hour.

(b) $P(t)=200e^{((\ln 2)t/4)}$.

(c) $P(6)=200e^{(3\ln 2/2)}=200\cdot 2\sqrt{2}=$**$400\sqrt{2}$**.

(d) $P'(4)=kP(4)=(\ln 2/4)(400)=$**$100\ln 2$ population units per hour**.

SCORING RUBRIC

Criterion	Points
Uses exponential model and solves $k=\ln 2/4$	3
Correct particular solution	2
Correct $P(6)=400\sqrt{2}$	2
Correct $P'(4)=100\ln 2$ with units	2
Total	9

Question 96: Radioactive Decay and Half-Life

★★★★★ Very Important

MODEL SOLUTION

(a) Half-life gives $e^{-8k}=1/2$, hence **$k=\ln 2/8$** per hour.
(b) **A(t)=640e^(-(ln2)t/8)**.
(c) Twenty-four hours is three half-lives, so $A(24)=640(1/2)^3=$**80 g**.
(d) $40/640=1/16=(1/2)^4$, so four half-lives are needed. Time=**32 hours**.

SCORING RUBRIC

Criterion	Points
Correct half-life equation and k	2
Correct exponential model	2
Correct mass 80 g	2
Recognizes four half-lives and finds 32 hours	3
Total	9

Question 97: Determining a Logistic Model from Rates

★★★★★ Very Important

MODEL SOLUTION

(a) The nonzero population where the rate is zero is the carrying capacity, so **K=200**.
(b) At P=50, 30=r(50)(1−50/200)=37.5r. Thus **r=0.8** per time unit.
(c) Logistic growth is maximal at P=K/2=100. From the table the maximum rate is **40 population units per time unit at P=100**.
(d) P=0 is unstable for positive populations because rates are positive just above 0. P=200 is stable because rates are positive below it and negative above it.

SCORING RUBRIC

Criterion	Points
Identifies K=200 from nonzero zero-rate equilibrium	2
Solves for r=0.8	2
Maximum rate 40 at P=100	3
Correct stability classification of both equilibria	2
Total	9

Question 98: Solving a Logistic Initial-Value Problem

★★★★★ Very Important

MODEL SOLUTION

(a) Setting $P'=0$ gives **$P=0$ and $P=200$**.

(b) Separating and integrating gives $\ln(P/(200-P))=0.4t+C$. Equivalently $P=200/(1+Ae^{-0.4t})$. $P(0)=50$ gives $A=3$, so **$P(t)=200/(1+3e^{-0.4t})$**.

(c) Set $P=100$: $2=1+3e^{-0.4t}$, so $e^{-0.4t}=1/3$ and **$t=\ln 3/0.4$**.

(d) As $t\to\infty$, $e^{-0.4t}\to 0$, so **$P(t)\to 200$**.

SCORING RUBRIC

Criterion	Points
Both equilibria	2
Separation/integration or correct logistic form	2
Uses initial condition to find $A=3$	1
Correct time $\ln 3/0.4$	2
Correct limit 200	2
Total	9

Question 99: Logistic Model for Adoption

★★★★★ Very Important

MODEL SOLUTION

(a) The equilibria are **A=0 and A=1000**; the carrying capacity is **1000 users**.
(b) $A(t)=1000/(1+Ce^{-0.6t})$. $A(0)=100$ gives $C=9$, so $\mathbf{A(t)=1000/(1+9e^{-0.6t})}$.
(c) Set $A=500$: $2=1+9e^{-0.6t}$. Thus $e^{-0.6t}=1/9$ and **$t=\ln 9/0.6$** time units.
(d) $A'(0)=0.6(100)(1-100/1000)=60(0.9)=$**54 users per time unit**.

SCORING RUBRIC

Criterion	Points
Equilibria and carrying capacity	2
Correct logistic form and constant C=9	2
Correct time ln9/0.6	3
Initial rate 54 with units	2
Total	9

Question 100: Newton's Cooling as a Differential Equation

★★★★★ Very Important

MODEL SOLUTION

(a) Equilibrium occurs when $T'=0$, giving **$T=20°C$**. This is the ambient room temperature toward which the liquid cools.
(b) The solution has form $T=20+Ce^{-kt}$. Since $T(0)=100$, $C=80$. At $t=2$, $60=20+80e^{-2k}$, so $e^{-2k}=1/2$ and **$k=\ln 2/2$** per minute.
(c) **T(t)=20+80e^(-(ln2)t/2)**. At $t=4$, $T(4)=20+80e^{-2\ln 2}=20+80/4=$**$40°C$**, matching the table.
(d) $T'(0)=-k(100-20)=-80(\ln 2/2)=$**$-40\ln 2$ °C per minute**. The negative sign means the temperature is decreasing.

SCORING RUBRIC

Criterion	Points
Equilibrium 20°C and contextual meaning	2
Uses table to find k=ln2/2	2
Correct model	1
Checks T(4)=40°C	2
Correct initial rate, sign, and units	2
Total	9

Core Differential-Equation Ideas

- **Exponential model:** $y'=ky$ gives $y=Ce^{kt}$. Positive k gives growth; negative k gives decay.
- **Equilibrium:** For $y'=f(y)$, constant solutions satisfy $f(y)=0$.
- **Slope field:** At each point (x,y), the segment slope is the value specified by $y'=f(x,y)$.
- **Separation:** Rearrange to place y terms with dy and x terms with dx, integrate, then apply initial data.
- **Logistic model:** $P'=rP(1-P/K)$, with equilibria 0 and K; K is the carrying capacity and maximum growth occurs at $P=K/2$.
- **Logistic solution:** $P(t)=K/(1+Ae^{-rt})$, where $A=K/P(0)-1$ for a positive initial population.

Disclaimer: Original AP Calculus AB-aligned practice. Not affiliated with, endorsed by, or approved by the College Board.